



"Navigating the Financial Engineer's Guide to Tesla Car Ownership and Budgeting"

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Abstract

This paper explores the financial considerations and budgeting strategies associated with buying a Tesla Model 3 for commuting purposes among students at San Jose State University. We examine the decision-making process behind selecting an electric vehicle over traditional gasoline-powered cars, focusing on the cost-saving potential, tax incentives, and environmental benefits. Through comparative analysis with other electric vehicle options and thorough financial modeling, we elucidate the long-term affordability and budgeting implications of owning a Tesla Model 3. Our findings highlight the financial advantages of electric vehicles, particularly in the context of rising gas prices and increasing environmental concerns. By choosing the Tesla Model 3, students can enjoy not only the convenience of electric transportation but also significant cost savings over the vehicle's lifespan. This paper provides valuable insights for students and individuals considering electric vehicle ownership in Silicon Valley and beyond. The method used involved assessing the initial purchase cost of a Tesla Model 3, evaluating affordability within a weekly income, and projecting monthly budget constraints. Additionally, fixed yearly operating and maintenance costs were considered.

Background of the problem

San Jose State University, also known as a commuter college, has various methods of transportation for students to commute to campus. There are discount train tickets, free buses, and light rail rides, but ultimately, driving to campus is the most convenient option. With a vehicle, students can set their schedules, carpool with friends, and have a smoother commute with time saved. For this project, we chose to pick a popular car, Tesla Model 3, due to its popularity in Silicon Valley, based on tax incentives, and potential fuel cost savings.

After narrowing down the options, purchasing a vehicle to commute to school and work was the problem we chose based on the variety of car options available ranging from old, new, or leasing a vehicle. With vehicle choices, we considered gas or electric vehicles and decided to go with electric due to the increasing popularity and rising gas prices. Some electric vehicle options include the Nissan Leaf, Hyundai Ioniq 6, Tesla Model 3, and Kia EV6. Comparing all the options, charging stations, incentives, reliability, perks, and technology in mind, Tesla was selected.

Alternative problem that you considered working on instead of this one

With the transition from a high school student to a full-fledged adult come major responsibilities. There are bills to pay, transportation considerations in getting from point A to point B, and affording basic necessities. Alternative problems that were considered was paying off student loan debt after college, lifestyle budgeting, and a first big purchase. With those ideas in mind, we decided to go with something more straightforward, a first big purchase. The first big purchase is

usually the first milestone into adulthood and it includes purchasing a house, adopting a pet, purchasing a car, or starting a business.

Method used to solve this problem

Starting the research process, the market price of the Tesla Model 3 started at \$36,730. As shown in Table 1, some fees factored into the final total amount included the DMV registration fee, license fee, order fee, vehicle trade-in, and tax. Adding all the costs together, the total came to \$40,991 (*"2023 Tesla Model 3 review, pricing, and Specs"*).

Table 1: The initial cost to purchase the vehicle

Items	Cost
Market price	36,730
DMV registration fee	\$297
Tax	9.375%
Tax (9.375%)	\$3,443
License fee	\$271
Order fee	\$250
Vehicle trade in	\$0
Total cost	\$40,992

The next step in the process was to understand the current lifestyle situation and if this purchase was within our budget and range. The independent variable for lifestyle affordability in this problem was the weekly \$4,000 per week income, with some wiggle room between food, rent, entertainment, and savings account/emergency fund as shown in Table 2. With income as a positive value and savings, food, rent, and entertainment as a cost, we are determining if purchasing the electric vehicle is feasible. The total monthly cost came out to \$2500 which shows that there is a \$1000 per month maximum for car payments, but to be cautious and prepared for emergencies and yearly car maintenance costs, the goal is for the car loan to be lower than \$800 a month.

Table 2: Monthly budget

Monthly budget	Cost/Value
Income	\$4,000
Saving account/emergency fund	500
Food	800
Rent	1000
Entertainment	200
Car payment	?
Total monthly fund/spending	2500

A factor that is commonly overlooked when purchasing a vehicle is the yearly operating and maintenance costs. These factors are subjective because they are based on the student's daily commute, whether they are safe and careful drivers, and how well they take care of their car. For this situation, all values in Table 3 are fixed. Assuming the student commutes around 280 miles a week, the cost to charge the vehicle is roughly \$20 per week which comes out to \$1,040 per year (Tingwall, 2024).

Table 3: Operation and Maintenance Cost

Average to charge car per week	\$20
Maintenance Cost (per year)	\$832
Insurance cost (per year)	\$2,052
Interest rate of loan	5.34%
Monthly Interest rate	0.45%
Loan length (years)	5

Results

Once understanding the budget and limitations per the student's lifestyle, with 5.34% interest, 60 months, and a starting cost of \$40,991 this problem was broken down by the amount of money paid towards interest and the principal. The

principal cost is the money being put towards paying off the loan, while the interest cost is the amount being paid for borrowing the money (Rivelli). When calculating the interest using Formula 1, the excel formula consisted of the interest 5.34% divided by 12 given there are 12 months in a year, the current payment month, the number of months the loan is set to complete, and the loan amount. For example, if it's the beginning of year 2, which is month 13, we can either use Formula 1 or the easier method, Formula 2, which is the current loan balance at the end of year 1 ending with \$33,642.37 multiplied by .45%, the monthly interest resulting \$149.71 as interest cost.

IPMT(interest, current month, number of loan month, loan amount) (1)

Interest cost = (Ending balance of the previous month)*(5.34%/12 months) (2)

The benefit of using Formula 2 is that the excel concept and inputs are the same as Formula 3. Taking year 4 as an example, with Formula 3 consisting of PPMT function, when inputting the monthly interest of .45%, month 4, 60 months for the total amount of months loaned, and the total cost of the vehicle of \$40,991 into excel, the principal amount comes up to \$605.55.

PPMT(interest, current month, number of loan month, loan amount) (3)

After calculating and understanding how interest and principal cost works, we figure out how much is the monthly payment needed to finish the car loan in 60 months using Formula 4. With Formula 4, inputting .45%, followed by 60 months and \$40,991, the monthly payment comes out to \$779.95.

PMT(Interest, number of months, loan amount) (4)

Table 4 shows the amount paid and balance at the end of each year. We can see we pay more interest upfront then slowly more of the principal cost is reducing the loan amount every year. The 1st to 2nd year when taking the ending balance of \$33,642.37 subtracted by the year 2 balance of \$25,891.57, this totals out to \$7,750.80 and increases through the years. At the end with year 4 subtracted with year 5 the total comes out to \$9,094.23.

Table 4: Cash Flow table

Cost	Year 1	Year 2	Year 3	Year 4	Year 5
Initial cost	40,991.00	\$33,642.37	\$25,891.57	\$17,716.59	\$9,094.23
Interest rate	5.34%	5.34%	5.34%	5.34%	5.34%
Monthly interest rate	0.45%	0.45%	0.45%	0.45%	0.45%
Loan months	60	60	60	60	60
Monthly payment	\$779.95	\$779.95	\$779.95	\$779.95	\$779.95
Total amount paid that year	\$7,348.63	\$7,750.80	\$8,174.97	\$8,622.36	\$9,094.23
Balance	\$33,642.37	\$25,891.57	\$17,716.59	\$9,094.23	\$0.00
Difference	\$7,750.80	\$8,174.97	\$8,622.36	\$9,094.23	

Table 5 models the initial cost of \$40,991.53 borrowed from the bank from year 0. Year 1 through 5 takes account of the monthly loan payment of \$779.95, insurance of \$2,052, and maintenance cost of \$832 added together. From years 6-10, the only items the student needs to pay are the maintenance and insurance costs since the loan has been paid off in year 5. Referencing “Tesla Model 3 depreciation”, it shows that the vehicle depreciates down to 52.11% in year 10. When year 10 comes around, the student decides to sell the car and the car resale value comes out to \$19,140.00 (“Tesla Model 3 depreciation”).

Table 5: 10-year cash flow

Year	Cashflow
0	\$40,991.53
1	-\$3,424.63
2	-\$3,826.80
3	-\$4,250.97
4	-\$4,698.36
5	-\$5,170.23
6	-\$2,884.00
7	-\$2,884.00
8	-\$2,884.00
9	-\$2,884.00
10	\$15,216.00

The internal rate of return (IRR) shows how we can gain or lose from our investments. The IRR as calculated by using the IRR Formula 5. Using excel, we inputted year 0 through year 10 into the IRR excel function and the result showed -8%. While negative IRR means that this car is not worth the investment, the understanding is that this vehicle is a consumable purchase and has no return value unless used for business purposes. When calculating the NPV using Formula 6, we used the interest of 5.34% and cash flow from year 0-10, resulting in 20,535.08. The cost of capital shows much is needed in order to justify the car purchase. To calculate the cost of capital it would be the year 10 ending value since the loan has been paid off and we can use the resale value as a down payment for another car. In this case, it would be \$15,216.

IRR= IRR(Cash Flow values) (5)

NPV= (interest, cash flow per year) (6)

Cost of capital = ending value for year 10

Table 6: Rate of Return, Net Present worth, and Cost of Capital

IRR	-8%
NPV	-\$20,535.08
Cost of capital	\$15,216.00

Discussion

The introduction sets the stage by highlighting the transportation challenges faced by students at San Jose State University and identifies driving as the most convenient option. It narrows down to the problem of purchasing a vehicle for commuting, specifically focusing on electric vehicles (EVs) due to their popularity and potential cost savings. The Tesla Model 3 is chosen as the vehicle of interest based on various factors such as tax incentives, fuel costs, and its popularity in Silicon Valley.

The results section likely provided detailed insights into the financial aspects of owning and budgeting for a Tesla Model 3, including initial costs, operating expenses, and potential savings compared to traditional gasoline-powered vehicles. In this discussion section, we will address the questions posed in the introduction and provide a structured conclusion to the report.

1. **Transportation Convenience:** The discussion begins by reaffirming the importance of convenience in transportation for students. While public transportation options are available, driving offers flexibility and time-saving benefits, aligning with the needs of busy students balancing school and work commitments.
2. **Vehicle Choice Considerations:** We address the decision-making process behind selecting an electric vehicle, emphasizing the increasing popularity and cost-saving potential compared to traditional gasoline cars. The Tesla Model 3 stands out among other EV options due to its charging infrastructure, incentives, reliability, and advanced technology features, making it a compelling choice for students in Silicon Valley.

3. Financial Implications: The main focus of the discussion is on the financial aspects of owning a Tesla Model 3. We delve into the initial costs of purchasing the vehicle, including any tax incentives or rebates available, as well as ongoing expenses such as electricity costs for charging compared to gasoline expenses for a conventional car.
4. Budgeting and Cost Savings: We explore how budgeting for a Tesla Model 3 aligns with the financial goals of students. By analyzing the potential savings in fuel costs over the vehicle's lifespan, factoring in the rising prices of gasoline, and considering any maintenance or repair differences between electric and gasoline cars, we provide insights into the long-term affordability of owning a Tesla Model 3.

Purchasing a Tesla can be a significant financial decision, especially if the price of the car is high relative to our current financial situation. To determine the monthly payment needed to pay off the car within our desired time frame of 5 years, careful financial planning and analysis are essential.

Firstly, it's important to assess our current financial standing, including your income, expenses, and savings. Understanding the cash flow and available funds will help us determine how much we can comfortably allocate toward a monthly car payment without straining our budget. Next, consider the total cost of the Tesla car we intend to purchase, including any additional fees, taxes, and registration costs. Subtracting any down payment or trade-in value we plan to use towards the purchase to determine the amount we need to finance through a loan.

Once we have the total loan amount, research current interest rates offered by various lenders to estimate the interest rate we may qualify for based on our creditworthiness. Use an online loan calculator or financial software to calculate the monthly payment required to pay off the loan in 5 years, considering the interest rate and loan term. A few of them are:

- Texas Instruments BA II Plus
- Microsoft Excel
- Quicken
- Personal Capital
- Mint

It's crucial to analyze how this monthly payment fits into your current budget and lifestyle. Considering whether the monthly payment aligns with our other financial obligations and goals, such as savings, investments, housing expenses, and discretionary spending. Furthermore, factor in potential additional costs associated with owning a Tesla, such as insurance premiums, maintenance, and charging expenses. These ongoing expenses can impact our monthly budget beyond just the loan payment.

In the end, we have to evaluate whether purchasing the Tesla car aligns with our long-term financial goals and priorities. Will the ownership of the car bring us significant value and satisfaction, or are there alternative transportation options that may better suit our budget and lifestyle?

Ultimately, by carefully assessing our financial situation, researching loan options, and considering the broader impact on our budget and lifestyle, we can make an informed decision about purchasing a Tesla car that supports our financial well-being and goals.

Appendix:

2023 Tesla Model 3 review, pricing, and Specs. (n.d.). <https://www.caranddriver.com/tesla/model-3-2023>

Rivelli, E. (n.d.). Does paying principal lower car payments?. <https://www.caranddriver.com/.https://www.caranddriver.com/auto-loans/a41868484/does-paying-principal-lower-monthly-car-payment/>

Tesla Model 3 depreciation. Car Edge. (n.d.). <https://caredge.com/tesla/model-3/depreciation>

Tingwall, E. (2024, March 6). How much does it cost to charge a Tesla?. MotorTrend. <https://www.motortrend.com/features/how-much-does-it-cost-to-charge-a-tesla/>

Excel table solved:

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Order fee	\$250
Vehicle trade in	\$0
Total cost	\$40,992

The total cost: the sum of all costs shown above. $\$36,730 + 297 + 3,443 + 271 + 250 = \$40,992$.

Table 2: Monthly budget

Monthly budget	Cost/Value
Income	\$4,000
Saving account/emergency fund	500
Food	800
Rent	1000
Entertainment	200
Car payment	?
Total monthly fund/spending	2500

Feasible car payment: $(4000 - 500 - 800 - 1000 - 200) = 2500$

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Loan months	60	60	60	60	60
Monthly payment	\$779.95	\$779.95	\$779.95	\$779.95	\$779.95
Total amount paid that year	\$7,348.63	\$7,750.80	\$8,174.97	\$8,622.36	\$9,094.23
Balance	\$33,642.37	\$25,891.57	\$17,716.59	\$9,094.23	\$0.00
Difference	\$7,750.80	\$8,174.97	\$8,622.36	\$9,094.23	

Monthly interest rate = $5.24\% / 12 = .45\%$

Monthly payment = $PMT(5.24\% / 12, 60, 40,991) = \779.95

Sum of month 1-12 interest rate + sum of month 1-12 principal rate = total amount paid that year

Table 5: 10-year cash flow

1	\$3,424.63
2	\$3,826.80
3	\$4,250.97
4	\$4,698.36
5	\$5,170.23
6	-\$2,884.00
7	-\$2,884.00
8	-\$2,884.00
9	-\$2,884.00
10	\$15,216.00

Year 0: vehicle cost, year 1-5: monthly car payment-insurance-car charging-maintenance, year 6-9:-(insurance+gas+maintenance), year 10:Resale value-(insurance-car charging+maintenance)

Table 6: Rate of Return, Net Present worth, and Cost of Capital

IRR	-8%
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Cost of capital	\$15,216.00

Excel functional: IRR(total of cash flow from year 0-10)

NPV= (5.34%, year 0-10 cash flow)

Cost of capital = value left after selling the car